# POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Compiler Design

Semester: Spring

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

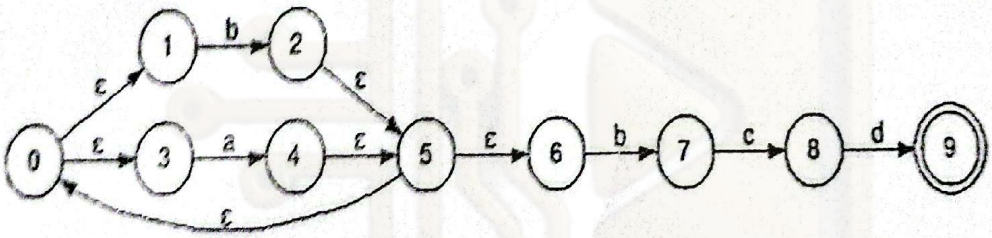
***Attempt all the questions.***

1. a) Define compiler. Explain the phases of a compiler with a detailed block diagram. 7

   b) Write code generation algorithm. Consider a statement x = (a-b)+(e+(c-d)) and perform simple code generation. 8

2. a) What are the functions of a lexical analyzer? Explain with suitable examples. How lexical analyzer use symbol table? 7

   b) Convert following NFA to DFA using Subset Construction. 8




3. a) Construct an **LR(1) parsing table** for the following grammar: 8

$S' \rightarrow S$

$S \rightarrow C\ C$

$C \rightarrow c\ C$

$C \rightarrow d$

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**OR**

Consider the following grammar for arithmetic expressions:

$E \rightarrow T\ E'$

$E' \rightarrow +\ T\ E' \mid \varepsilon$

$T \rightarrow F\ T'$

$T' \rightarrow *\ F\ T' \mid \varepsilon$

$F \rightarrow (\ E\ ) \mid id$

Construct the LL(1) parsing table.



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b) How does Recursive Descent Parsing works? Consider the following grammar 7

S → cAd

A → ab|a

Parse the string ***cad*** using recursive descent parser.

4. a) What is the significance of symbol tables in compilers? Explain the role of hash tables in building a symbol table. 7

b) What is the need of intermediate code optimization? Discuss the different types of Graphical IRs used in compilers. 8

5. a) Discuss the concept of Register Allocation and Assignment in code generation. 8

**OR**

Explain the major issues in the design of a code generator.

b) Convert regular expression (a|b)*abb to NFA using Thompson's Construction and then perform NFA to DFA conversion. 7

6. a) What are the principal sources of optimization in compiler design? Explain with examples. 7

b) Discuss Data-Flow Analysis in detail. How are Data-Flow Equations solved? 8

7. Write short notes on: (**Any two**) 2×5

a) Mapping Values to Names

b) Control Flow Graph

c) Closure properties of REs

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